

DAY 179/365 • SVM • PART 3 OF 4

# When no straight line works: the kernel trick

Bending the problem instead of the boundary.

# A problem a straight line can't solve

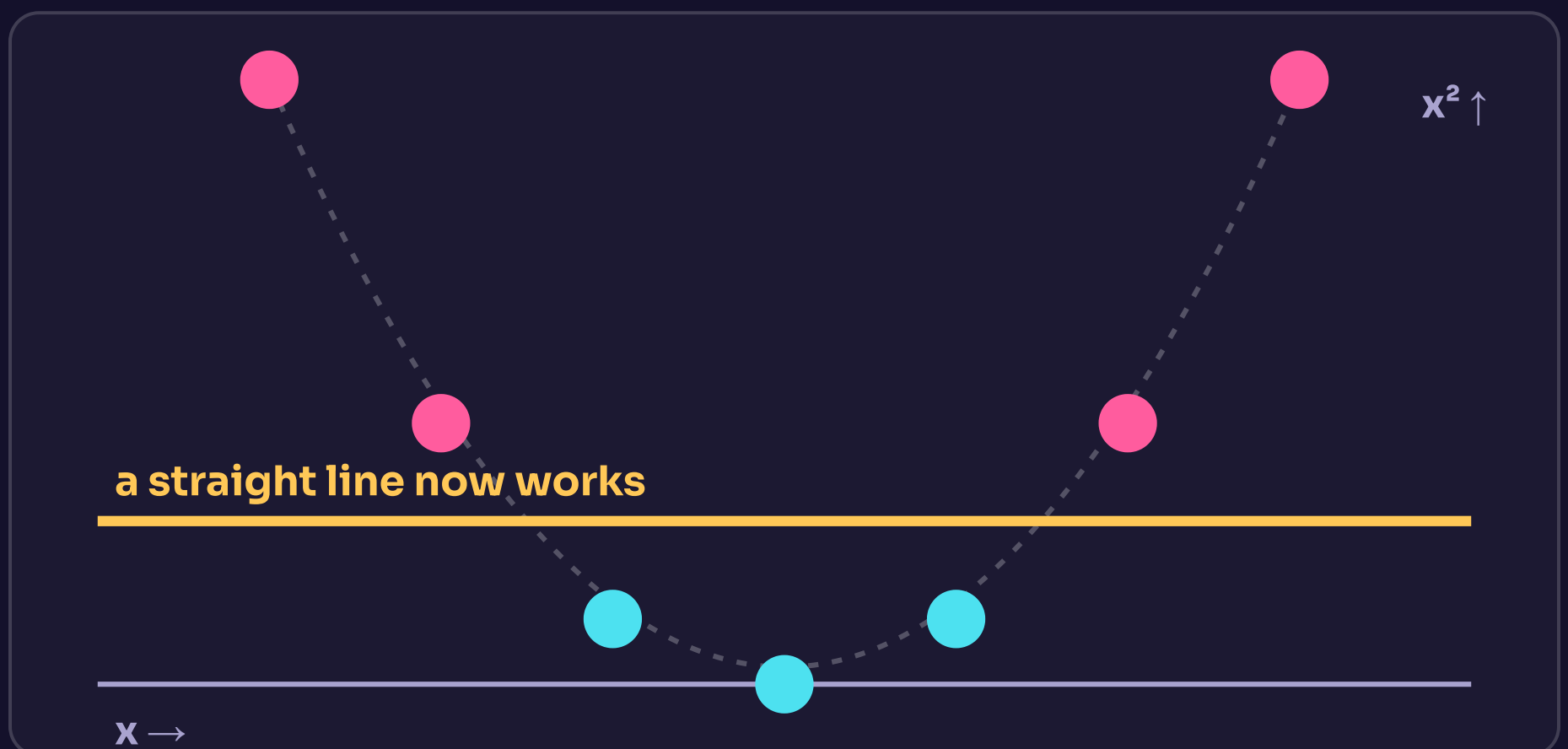
Parts 1 and 2 drew a straight road between two groups. But look at this data on a single number line:



Pink sits on **both sides** of cyan. A single cut point can't separate them. Two cuts could, but SVM's boundary is a single straight one.

# The fix: lift the points up

Add a second number to every point: its position squared. So the point  $x = -3$  becomes  $(-3, 9)$ .



The cyan points sit low ( $x^2$  of 0 or 1). The pink points sit high ( $x^2$  of 4 or 9). A flat line at height 2.5 splits them cleanly.

# You've seen this move before

## Day 167: polynomial regression

The straight-line model couldn't bend, so I added  $x^2$  as an extra column. Same model, richer input.

## Today: same trick for classification

A straight boundary couldn't separate the groups, so we add  $x^2$  and draw the line in the new space. Back in the original space, that flat line becomes **two cut points**, a curved boundary.

Wherever the lifted data can be split by a straight line, the original data gets a boundary that bends.

**The catch:** with many features and higher powers, writing out all the new columns gets huge. That's where the "trick" comes in.

# The kernel trick: skip the lifting

SVM only ever needs one thing from pairs of points: **how similar are they?** A **kernel** is a function that gives that similarity as if the points had been lifted, **without actually lifting them.**

## Worked example

Take  $a = (1, 2)$  and  $b = (3, 1)$ , and the kernel  $(a \cdot b + 1)^2$ .

| Way                              | Work                                      | Result    |
|----------------------------------|-------------------------------------------|-----------|
| Shortcut (kernel)                | $a \cdot b = 5$ , then $(5 + 1)^2$        | <b>36</b> |
| Lift to 6 numbers, then multiply | 6 new columns each, then sum the products | <b>36</b> |

Same answer, but the shortcut never builds the 6 columns. With hundreds of features, that saving is the whole point.

Worked from the formula, using the standard 6-column expansion for 2 features.

# Three kernels to know

| Kernel     | Use it when                             | Boundary        |
|------------|-----------------------------------------|-----------------|
| Linear     | A straight line already works           | Straight        |
| Polynomial | Curves from powers, like the $x^2$ lift | Smooth curves   |
| RBF        | Complex, blobby patterns                | Flexible shapes |

## RBF in plain words

It measures similarity by **distance**. Close points score near 1, far points near 0.

$K = \exp(-\gamma \times \text{distance}^2)$ , with  $\gamma = 0.5$ :

Points (1, 2) and (1.5, 2):  $\text{distance}^2 = 0.25$ , so  $K \approx 0.88$  (very similar).

Points (1, 2) and (3, 1):  $\text{distance}^2 = 5$ , so  $K \approx 0.08$  (very different).

# Choosing and tuning in practice

## Start simple

Try a linear kernel first. If the score is poor and the data is curved, try RBF. In scikit-learn, RBF is the default for SVC.

## Two dials, tuned together

**C** from Part 2, and for RBF a second dial, **gamma**. A high gamma makes each point's influence tiny and local, so the boundary can wrap tightly around individual points and overfit.

## Scale your features

Kernels are built on distances, so a column in big units drowns out the rest. The article names scaling as a key weakness. Same lesson as my Day 161 standardize toggle.

Gamma and the "RBF is the default" point are standard scikit-learn practice, my addition to the article.

## TAKEAWAYS

**Can't separate it in 1D? Lift it to 2D and a line works.**

**Kernel trick:** get the similarity of lifted points without ever lifting them.

**Linear, polynomial, RBF:** start linear, go RBF for complex shapes, tune C and gamma together, and scale first.

**Part 4:** SVM in Python on the breast cancer data, plus advantages and limitations.